ML-DLOps Assignment-4

Optimizing Transformer Translation with Ray Tune Optuna

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GitHub Repository

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1. Baseline Performance

Before initiating the hyperparameter optimization sweep, the unoptimized base model was trained for 100 epochs using the predefined set of parameters. This established the functional baseline requirements.

Total Training Time: 121 minutes 55 seconds (˜122 mins)
Final Training Loss: 0.0984
Final BLEU Score: 56.65 (0.5665)

These initial results functioned as the benchmark, revealing that while the baseline achieved sufficient translation quality (BLEU > 50), it required a highly inefficient 100 epochs and nearly two hours to converge.

2. Hyperparameter Search Space Configuration

To improve convergence speed and model efficiency, five separate Transformer hyperparameters were tracked, heavily increasing the dimensions of the initial sweep. The optimization was executed using **Ray Tune** coupled with an **OptunaSearch** underlying algorithm prioritizing minimization of the validation `loss`.

The configured search space was strictly defined as follows:

Learning Rate (`lr`): Log-uniform scale (`1e-5` to `1e-3`)
Impact: Controls step sizes ensuring rapid but stable local minima convergence.
Batch Size (`batch_size`): Discrete categorical choice (`16, 32, 64`)
Impact: Directly affects memory mapping limit and stochastic gradient stability.
Attention Heads (`num_heads`): Discrete choice (`4, 8`)
Impact: Modifies representative subspaces holding $D_{model} = 512$ *strictly divisible.*
FeedForward Dimension (`d_ff`): Discrete categorical choice (`1024, 2048`)
Impact: Non-linear mapping capacity per encoder/decoder sublayer. **Dropout Rate (`dropout`):** Uniform continuous scale (`0.1` to `0.4`)
Impact: Generalizes parameter nodes preventing heavy dataset overfitting.

Coupled directly with this parameter space setup natively rested an **ASHAScheduler** limiting each trial to a maximum threshold of 20 epochs, strictly halting trailing branches performing uncompetitively to save parallel computation instances.

3. Best Model Configuration & Final Metrics

Out of the rigidly distributed 20 samples utilizing the Optuna metric-based reduction logic, the most dominant set of arguments arose reliably:

Best Trial Configuration Found:

```
{
    'lr': 7.710214593871093e-05,
    'batch_size': 16,
    'num_heads': 8,
    'd_ff': 1024,
    'dropout': 0.10497456049175113
}
```

Using this optimized configuration, a dedicated **Final Evaluation** training sweep bypassed the limitations capping the search bounds by training out for an elevated 30 epochs:

Total Training Time (30 Epochs): 35 minutes
Final Validation Loss: 0.1907
Final BLEU Score: 79.91 (0.7990)

4. Conclusion

The hyperparameter optimization effectively refactored the translation framework's computational efficiency structure natively. The optimized model dramatically **exceeded** the baseline BLEU score requirement significantly (+23.26 improvement) arriving efficiently at **79.91**. Additionally, the fine-tuned iteration completely eclipsed the baseline timeline, arriving favorably in merely **35 minutes over 30 epochs** in vast contrast to the 122-minute, 100-epoch baseline. This formally achieves the 'Efficiency Challenge" via the integration of robust ASHA trial schedulers prioritizing successful node retention dynamically computing `min(loss)`.